

$$m_1^2 \rightarrow -C_{\phi 1^2} + \frac{1}{16 \pi^2}$$

$$\begin{aligned} & \left(-\frac{1}{4} C_{\phi 1^2} (g_1^2 + 3 g_2^2) - C_{\phi 1^2} \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} \text{LF}_{2,1,0} [m_{\tilde{l}}^{\text{p}}, m_{\tilde{e}}^{\text{r}}] + C_{\phi 1^2} \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} \text{LF}_{3,1,-1} [m_{\tilde{l}}^{\text{p}}, m_{\tilde{e}}^{\text{r}}] - \right. \\ & C_{\phi 1^2}^2 \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} \text{LF}_{3,1,0} [m_{\tilde{l}}^{\text{p}}, m_{\tilde{e}}^{\text{r}}] + 3 C_{\phi 1^2}^2 \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} \text{LF}_{4,1,-1} [m_{\tilde{l}}^{\text{p}}, m_{\tilde{e}}^{\text{r}}] - \\ & 2 C_{\phi 1^2}^2 \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} \text{LF}_{5,1,-2} [m_{\tilde{l}}^{\text{p}}, m_{\tilde{e}}^{\text{r}}] - 3 C_{\phi 1^2} \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} \text{LF}_{2,1,0} [m_{\tilde{q}}^{\text{p}}, m_{\tilde{d}}^{\text{r}}] + \\ & 3 C_{\phi 1^2} \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} \text{LF}_{3,1,-1} [m_{\tilde{q}}^{\text{p}}, m_{\tilde{d}}^{\text{r}}] - 3 C_{\phi 1^2}^2 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} \text{LF}_{3,1,0} [m_{\tilde{q}}^{\text{p}}, m_{\tilde{d}}^{\text{r}}] + \\ & 9 C_{\phi 1^2}^2 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} \text{LF}_{4,1,-1} [m_{\tilde{q}}^{\text{p}}, m_{\tilde{d}}^{\text{r}}] - 6 C_{\phi 1^2}^2 \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} \text{LF}_{5,1,-2} [m_{\tilde{q}}^{\text{p}}, m_{\tilde{d}}^{\text{r}}] - \\ & 3 C_{\phi 1^2} \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{2,1,0} [m_{\tilde{u}}^{\text{r}}, m_{\tilde{q}}^{\text{p}}] + 3 C_{\phi 1^2} \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{3,1,-1} [m_{\tilde{u}}^{\text{r}}, m_{\tilde{q}}^{\text{p}}] - \\ & 3 C_{\phi 1^2}^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{3,1,0} [m_{\tilde{u}}^{\text{r}}, m_{\tilde{q}}^{\text{p}}] + 9 C_{\phi 1^2}^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{4,1,-1} [m_{\tilde{u}}^{\text{r}}, m_{\tilde{q}}^{\text{p}}] - \\ & 6 C_{\phi 1^2}^2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} \text{LF}_{5,1,-2} [m_{\tilde{u}}^{\text{r}}, m_{\tilde{q}}^{\text{p}}] + \frac{3}{2} C_{\phi 1^2} g_1^2 \text{LF}_{2,1,-1} [\tilde{\mu}, m_1] - \\ & C_{\phi 1^2} g_1^2 \text{LF}_{3,1,-2} [\tilde{\mu}, m_1] + 2 g_1^2 C_{\phi 1^2}^2 \text{LF}_{3,1,-1} [\tilde{\mu}, m_1] - 4 g_1^2 C_{\phi 1^2}^2 \text{LF}_{4,1,-2} [\tilde{\mu}, m_1] + \\ & 2 g_1^2 C_{\phi 1^2}^2 \text{LF}_{5,1,-3} [\tilde{\mu}, m_1] + \frac{9}{2} C_{\phi 1^2} g_2^2 \text{LF}_{2,1,-1} [\tilde{\mu}, m_2] - 3 C_{\phi 1^2} g_2^2 \text{LF}_{3,1,-2} [\tilde{\mu}, m_2] + \\ & \left. 6 g_2^2 C_{\phi 1^2}^2 \text{LF}_{3,1,-1} [\tilde{\mu}, m_2] - 12 g_2^2 C_{\phi 1^2}^2 \text{LF}_{4,1,-2} [\tilde{\mu}, m_2] + 6 g_2^2 C_{\phi 1^2}^2 \text{LF}_{5,1,-3} [\tilde{\mu}, m_2] \right) \end{aligned}$$